Why the three-body problem has no general closed-form solution

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We review the results that delimit the Newtonian three-body problem. The state of the system has 18 components, and the symmetries provide only 10 classical first integrals. The theorems of Bruns (1887) and Poincaré (1890) exclude additional algebraic and uniform analytic integrals, but not other representations. Sundman's series (1912) solves the problem for nonzero total angular momentum, with convergence too slow for practical use. In the nonhierarchical regime, the exponential divergence of trajectories makes the predictability horizon grow only with the logarithm of the initial precision, a limit that no exact formula would remove. We conclude that the open question is not that of a closed-form formula, but rather that of understanding the structures that make the motion regular and of describing the chaotic regime statistically, and we indicate the corresponding research directions.

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I. INTRODUCTION

The three-body problem asks how three point masses interacting only through Newtonian gravitation evolve in time, given their initial positions and velocities. For two bodies, the answer has been complete since Newton: the relative orbits are conic sections, and the position at any instant follows from Kepler's equation. For three bodies, no general solution expressed in elementary functions exists, and the reason is structural, not a lack of ingenuity.

This note gathers, in compact form, the results that delimit the problem: the counting of first integrals, the impossibility theorems of Bruns and Poincaré [1, 2], the convergent series of Sundman [3], and the predictability limit imposed by chaos. Our aim is to separate precisely what has been proven from what remains open, as a basis for the subsequent notes in the series.

II. FORMULATION

With masses m_i and positions $\mathbf{r}_i$, $i = 1,2,3$, the equations of motion are

$$\ddot{\mathbf{r}}_i = \sum_{j\neq i} G m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}. \tag{1}$$

In three-dimensional space, the state of the system has 18 components (three of position and three of velocity per body). The symmetries of the problem provide 10 classical first integrals: energy (1), linear momentum and uniform motion of the center of mass (6), and angular momentum (3). In the two-body problem, these integrals, together with the Laplace-Runge-Lenz vector, suffice to reduce the motion to quadratures. In the three-body problem, degrees of freedom remain without a corresponding integral.

III. IMPOSSIBILITY RESULTS

In 1887, Bruns proved that the problem admits no new first integrals that are algebraic functions of the Cartesian coordinates and velocities, beyond the ten classical ones [1]. Poincaré extended the result to uniform analytic integrals in the restricted problem and, in doing so, described for the first time the tangling of stable and unstable manifolds that today characterizes deterministic chaos [2].

It is important to delimit the scope of these theorems. They exclude a precise class of solutions, built from a precise class of functions. They do not exclude representations by other functions, just as the impossibility of solving the general quintic equation by radicals (Abel, 1824) did not prevent its solution by elliptic functions (Hermite, 1858).

IV. THE SUNDMAN SERIES

In 1912, Sundman obtained a general solution, valid for all initial conditions with nonzero total angular momentum, in the form of power series in a regularized time variable [3]. The regularization handles binary collisions, which are the singularities of the problem, and makes the series convergent for all time. The result is mathematically complete, but its convergence is so slow that later estimates indicated that a number of terms of order $10^{8\,000\,000}$ would be needed for precision useful in astronomy [7]. From a practical standpoint, the series does not replace numerical integration.

V. SENSITIVITY TO INITIAL CONDITIONS

The decisive practical difficulty lies elsewhere. In the nonhierarchical regime, that is, when the three masses are comparable and approach one another, the system is chaotic: two initial conditions separated by δ_0 diverge, on average, as

$$\delta(t) \sim \delta_0 e^{\lambda t}, \tag{2}$$

where λ is the largest Lyapunov exponent. The predictability horizon grows only with the logarithm of the initial precision: each additional correct digit in the initial conditions adds a fixed interval, of order $\ln 10/\lambda$, to the prediction time. An exact solution, even if it existed in usable form, would not remove this limit, because initial conditions are always known with finite precision.

VI. CURRENT RESEARCH DIRECTIONS

In view of this, contemporary research is organized along directions that do not depend on a closed-form formula:

- **Exact particular solutions:** the collinear configurations of Euler (1767), the equilateral triangle of Lagrange (1772), and periodic orbits such as the equal-mass figure eight, proven by Chenciner and Montgomery [4]. Numerical searches since 2013 have catalogued hundreds and then thousands of periodic orbits [6].
- **Stability criteria:** conditions under which particular configurations or hierarchical systems remain regular over long times.
- **Statistical descriptions:** in the chaotic regime, predicting the distribution of outcomes, typically the formation of a binary and the ejection of the third body, rather than the individual trajectory [5].
- **Geometry of shape space:** reduction by symmetries maps the motion onto a shape sphere, on which alignments and collisions occupy definite locations and trajectories can be encoded symbolically [8].

VII. CONCLUSIONS

What has been proven is the nonexistence of new algebraic or uniform analytic integrals, the existence of a convergent series without practical utility, and the exponential sensitivity to initial conditions in the nonhierarchical regime. What remains open is not a formula, but rather the understanding of which structures make the motion regular and of how to describe statistically what is chaotic. The subsequent notes in this series report numerical experiments in this direction, in particular on the mechanisms that make few-body configurations stable and on intuitive stabilization proposals that fail.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

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